

Exact Paper Model Bank Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the exact-paper model-bank campaign prepared in:

- `doc/reports/campaign_plans/track_1/exact_paper/2026-04-10-17-04-41_exact_paper_model_bank_campaign_plan_report.md`

The campaign executed 4 repository-owned validation runs through the dedicated exact-paper launcher:

- completed runs: 4
- failed runs: 0
- execution window: `2026-04-10T18:44:39.0700428+02:00` to `2026-04-10T19:14:18.3733939+02:00`
- campaign artifact root:
`output/training_campaigns/track1/exact_paper/forward/uncategorized/shared/exact_paper_model_bank_campaign_2026_04_10_17_04_41/` Unlike the harmonic-wise RCIM Model-Bank Reproduction campaign, this package does not optimize the full TE-curve offline benchmark directly. Its purpose is to validate the stricter recovered-paper branch that uses:
 - recovered `rpm`, `deg`, `tor` inputs;
 - exact `ampl_k` and `phase_k` targets for harmonics
`0, 1, 3, 39, 40, 78, 81, 156, 162, 240` ;
 - the full recovered family bank;
 - one ONNX export artifact per family and target.

Objective And Outcome

The campaign had four concrete questions

1. can the full bank finish in diagnostic mode without ONNX export crashes?
2. can the same bank finish in strict mode after the HGBM converter fix?
3. does the isolated SVR branch still need surrogate safeguards, and does it now serialize cleanly?
4. can the non-SVR branch export cleanly as a reference bank?

Outcome:

- the full exact-paper bank now completes in both diagnostic continue mode and strict reference mode;
- the HGBM ONNX export blocker is resolved under the current local stack;
- the isolated SVR branch exports all 20 targets successfully, with 5 constant-surrogate replacements still required for degenerate targets;
- the non-SVR branch exports its full 180 -file surface cleanly;
- the canonical winner is the strict full-bank reference run
exact_full_bank_strict_reference .

Ranking Policy

This campaign is a stabilization campaign, not a generic hyperparameter sweep. The ranking therefore follows this explicit serialized policy:

- primary metric: winning_mean_component_mape_percent
- first tie breaker: winning_mean_component_mae
- second tie breaker: prefer strict export mode over continue
- third tie breaker: prefer broader enabled-family coverage
- fourth tie breaker: winning_mean_component_rmse
- fifth tie breaker: lexical run_name

That policy intentionally promotes the cleanest faithful reference when the diagnostic and strict runs produce equivalent validation metrics.

Campaign Ranking

Ranked Completed Runs

Rank	Config	Family Scope	Export Mode	Winner	Mean Component MAPE [%]
1	exact_full_bank_strict_reference	all 10 recovered families	strict	RF	18.369
2	exact_full_bank_diagnostic_continue	all 10 recovered families	continue	RF	18.369
3	exact_non_svr_export_reference	9 non- SVR families	strict	RF	18.369
4	exact_svr_export_diagnostic	SVR only	continue	SVR	103.265

Config	Mean Component MAE	ONNX Exported	Surrogates	Failed Exports
exact_full_bank_strict_reference	0.056284	200	5	0
exact_full_bank_diagnostic_continue	0.056284	200	5	0
exact_non_svr_export_reference	0.056284	180	0	0
exact_svr_export_diagnostic	0.138346	20	5	0

Campaign Winner

The explicit campaign winner is

- exact_full_bank_strict_reference

Its result was:

- winner family: RF
- winner family estimator: RandomForestRegressor
- enabled families: SVR, MLP, RF, DT, ET, ERT, GBM, HGBM, XGBM, LGBM
- winner mean component MAPE: 18.369%
- winner mean component MAE: 0.056284
- winner mean component RMSE: 0.144839
- ONNX exported files: 200
- failed exports: 0
- surrogate exports: 5

This run won because it matched the best achieved component metric while also proving that the full exact-paper bank now survives strict export mode without fallback continuation behavior.

Top Family Ranking Inside The Winning Run

Rank	Family	Mean Component MAPE [%]	Mean Component MAE	Mean Component RMSE
1	RF	18.369	0.056284	0.144839
2	HGBM	20.586	0.079797	0.155340
3	ERT	21.017	0.054424	0.139747
4	GBM	24.343	0.062593	0.152818
5	DT	27.419	0.063643	0.170755

Interpretation By Experiment Block

1. The Full Strict Reference Is Now Operational

The most important campaign result is operational rather than purely numerical. The full strict exact-paper bank now completes end to end:

- all 10 recovered families fitted;
- all 20 harmonic targets evaluated per family;
- all 200 per-target ONNX artifacts exported;
- zero export failures under strict mode.

Interpretation:

- the exact-paper branch is no longer only prepared;
- the HGBM ONNX export blocker has been removed in the repository-owned support layer;
- the campaign now provides a stable exact-paper reference surface for future comparison and deployment-facing inspection.

2. The Diagnostic Continue Run Is Now A Safety Net, Not The Mainline

The full-bank diagnostic run now reaches the same winner metric as the strict reference:

- full diagnostic winner: RF at 18.369%
- full strict winner: RF at 18.369%

Interpretation:

- diagnostic mode remains useful for future isolation work;
- however, the branch no longer depends on diagnostic continuation to finish;
- promotion should therefore move to the strict full-bank run, not stay on the continue-mode path.

3. The SVR Branch Is Stable, But It Still Requires Surrogate Handling

The isolated SVR export diagnostic completed successfully:

- exported targets: 20
- failed exports: 0
- constant-surrogate exports: 5

Interpretation:

- the earlier degenerate- SVR converter crash is resolved operationally;
- the branch is still not a pure raw- SVR export path, because 5 targets remain constant predictors under the modern stack;
- the surrogate behavior is now a documented compatibility layer rather than a hidden failure.

4. The Non- SVR Branch Confirms The Remaining Bank Is Clean

The non- SVR reference run exported the remaining family bank independently:

- enabled families: 9
- exported files: 180
- failed exports: 0

Interpretation:

- the rest of the recovered family bank no longer depends on the SVR safeguard to remain operational;
- the branch can now be discussed as two stable pieces:
 - the special-case SVR branch with documented surrogate handling;
 - the non- SVR branch with clean strict export behavior.

Error And Family Findings

The winning strict full-bank run keeps the same best family ordering as the diagnostic full-bank run. The best exact-paper family in this repository branch is still:

- RF

The highest-contributing winning-family target errors remain concentrated in the phase branch:

- fft_y_Fw_filtered_phase_156 MAE 0.425070
- fft_y_Fw_filtered_phase_240 MAE 0.277042
- fft_y_Fw_filtered_phase_162 MAE 0.230457
- fft_y_Fw_filtered_phase_78 MAE 0.051585
- fft_y_Fw_filtered_phase_81 MAE 0.047979

Interpretation:

- the exact-paper branch remains dominated by phase-target difficulty rather than amplitude-target difficulty;
- the exact family-bank workflow is now stable enough that future work can focus on interpretation and parity analysis instead of export rescue.

ONNX Export Alignment Findings

The campaign succeeded operationally, but the path-level recovered-reference comparison remains intentionally unresolved:

- full-bank exported files: 200
- recovered reference files: 201
- matched relative paths: 0
- missing relative paths against reference: 201
- extra exported relative paths: 200

This does **not** mean the export failed. The mismatch is structural:

- the recovered paper bundle uses nested family-plus-kind paths such as
DT/amp1/DecisionTreeRegressor_amp10.onnx ;
- the repository export currently writes flat family folders such as
DT/DecisionTreeRegressor_amp10.onnx ;
- the recovered bundle also contains at least one duplicate-style filename
variant such as RF/amp1/RandomForestRegressor_amp1240 (1).onnx . Interpretation:
- exportability is now stable;
- semantic path parity with the recovered release is still open;
- the next exact-paper refinement should normalize repository export layout if
strict recovered-path mirroring is required.

Paper Alignment Impact

This campaign materially changes the exact-paper branch in three ways.

What Improved

- the exact-paper model-bank launcher is no longer only campaign-prepared;
- the full strict recovered family bank now executes end to end;
- the SVR and HGBM export blockers are both operationally contained.

What Remains Open

- the exact branch still uses component-level family metrics, not the harmonic reconstruction TE-curve benchmark used for Target A ;
- recovered-path ONNX parity is still not normalized;
- no online compensation benchmark exists yet.

What The Campaign Established

- the exact-paper branch is now a stable structural reference bank;
- the harmonic-wise branch remains the current Target A offline benchmark;
- future paper-comparison reporting can now distinguish clearly between:
 - harmonic-wise TE-curve benchmark status;
 - exact family-bank export and parity status.

Main Conclusions

The campaign supports five conclusions.

1. The Exact-Paper Branch Is Now Campaign-Executable

The repository can now run the full exact-paper family bank through the canonical batch launcher without manual intervention.

2. The Correct Promoted Winner Is The Strict Full-Bank Reference

The diagnostic full-bank run and the strict reference reached the same winning metric, so promotion should prefer the strict exact-paper reference because it proves the branch no longer needs diagnostic continuation behavior.

3. The HGBM Export Issue Was Real And Is Now Resolved

The campaign would not have finished without the repository-side HGBM converter sanitation fix. That fix is now validated by the real four-run batch.

4. SVR Is Operationally Stable But Still Special-Case

The branch is exportable again, but only because the repository now documents and applies constant-surrogate handling for degenerate SVR targets.

5. Exact-Bank Success Does Not Yet Close Target A

This campaign stabilizes the exact-paper branch, but it does not replace the harmonic-wise offline TE-curve benchmark that still sits at 8.877% and remains above the paper's 4.7% threshold.

Recommended Next Actions

The next technically justified steps are

1. keep exact_full_bank_strict_reference as the canonical exact-paper run when discussing recovered-family-bank status;
2. preserve the SVR diagnostic branch for future surrogate-audit work rather than as the promoted reference;
3. decide whether the next exact-paper task should target recovered ONNX path mirroring, because the current export surface is operationally valid but not path-identical to the recovered release;
4. keep Target A tracking anchored to the harmonic-wise offline branch until the exact bank is connected to a TE-curve reconstruction benchmark;
5. defer any LAN-remote rerun unless future exact-bank iterations become materially heavier than the current local stabilization pass.

Artifact References

- Campaign root:
`output/training_campaigns/track1/exact_paper/forward/uncategorized/shared/exact_paper_model_bank_campaign_2026_04_10_17_04_41/`
- Campaign leaderboard:
`output/training_campaigns/track1/exact_paper/forward/uncategorized/shared/exact_paper_model_bank_campaign_2026_04_10_17_04_41/campaign_leaderboard.yaml`
- Campaign best run YAML:
`output/training_campaigns/track1/exact_paper/forward/uncategorized/shared/exact_paper_model_bank_campaign_2026_04_10_17_04_41/campaign_best_run.yaml`
- Campaign best run note:
`output/training_campaigns/track1/exact_paper/forward/uncategorized/shared/exact_paper_model_bank_campaign_2026_04_10_17_04_41/campaign_best_run.md`
- Winning strict validation summary:
`output/validation_checks/paper_reimplementation_rcim_exact_model_bank/forward/baseline_reproduction/shared/2026-04-10-19-12-14__exact_full_bank_strict_reference_campaign_run/validation_summary.yaml`
- Winning strict model bundle:
`output/validation_checks/paper_reimplementation_rcim_exact_model_bank/forward/baseline_reproduction/shared/2026-04-10-19-12-14__exact_full_bank_strict_reference_campaign_run/paper_family_model_bank.pkl`
- Diagnostic SVR validation summary:
`output/validation_checks/paper_reimplementation_rcim_exact_model_bank/forward/baseline_reproduction/shared/2026-04-10-19-12-57__exact_svr_export_diagnostic_campaign_run/validation_summary.yaml`
- Per-run launcher logs:
`output/training_campaigns/track1/exact_paper/forward/uncategorized/shared/exact_paper_model_bank_campaign_2026_04_10_17_04_41/logs/`